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Field evaluation of particulate matter measurements using tapered element oscillating microbalance in a layer house

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The tapered element oscillating microbalance (TEOM) is one type of continuous ambient particulate matter (PM) monitor. Adsorption and desorption of moisture and semivolatile species may cause positive or negative artifacts in TEOM PM mass measurement. The objective of this field study was to investigate possible uncertainties associated with TEOM measurements in the poultry operation environment. For comparisons of TEOM with filter-based gravimetric method, four instruments (TEOM-PM₁₀, low-volume PM₁₀ sampler, TEOM-PM_{2.5}, and PM_{2.5} speciation sampler) were collocated and tested inside a poultry house for PM_{2.5} and PM₁₀ (PM with aerodynamic equivalent diameter ≤ 2.5 and ≤ 10 μm , respectively) measurements. Fifteen sets of 24-hr PM₁₀ concentrations and 13 sets of 24-hr PM_{2.5} measurements were obtained. Results indicate that compared with filter-based gravimetric method, TEOM gave significantly lower values of both PM₁₀ and PM_{2.5} mass concentrations. For PM₁₀, the average ratio of TEOM to the gravimetric method was 0.936. For PM_{2.5}, the average ratio of TEOM to the gravimetric method was 0.738. Particulate matter in the poultry houses possibly contains semivolatile compounds and moisture due to high levels of relative humidity (RH) and gas pollutants. The internal heating mechanism of the TEOM may cause losses in mass through volatilization. To investigate the effects of TEOM settings on concentration measurements, the heaters of two identical TEOMs were set at 50 °C, 30 °C, or no heating at all. They were collocated and tested for total suspended particle (TSP), PM₁₀, and PM_{2.5} measurements in layer house for 6 weeks. For all TSP, PM₁₀, and PM_{2.5} measurements, the internal TEOM temperature setting had a significant effect ($P < 0.05$). Significantly higher PM mass concentrations were measured at lower temperature settings. The effects of environmental (i.e., temperature, RH, NH₃ and CO₂ concentrations) and instrumental (i.e., filter loading and noise) parameters on PM measurements were also assessed using regression analysis.

Implications: Because of its potential health and environmental effects, particulate matter (PM) emissions from animal feeding operations (AFOs) have been a great concern to the public and to the regulatory agencies. The tapered element oscillating microbalance (TEOM) PM monitor has been adapted for continuous PM measurements in some AFO air quality studies. This study investigated possible uncertainties associated with TEOM measurements in an egg production environment. It was discovered that there was a significant bias in TEOM measurements of PM₁₀ as compared with federal reference method. Internal temperature settings of a TEOM have significant impact on its PM measurement.

Introduction

As a US Environmental Protection Agency (EPA) federal equivalent method (FEM) for particulate matter with aerodynamic equivalent diameter ≤ 10 μm (PM₁₀) measurements (US Environmental Protection Agency, 1997), the tapered element oscillating microbalance (TEOM) is a continuous ambient PM monitor. The TEOM has been widely used in a variety of air quality studies and monitoring in recent years (Gilbert and Clark, 2001; Green and Fuller, 2006; Grover et al., 2005; Rogers et al., 1998; Schwab et al., 2004). The TEOM measures PM mass concentration continuously with a microbalance. The filter and the sampled air passing through filter are conditioned to a

constant temperature (50 or 30 °C) to minimize interference of water condensation and temperature variations with mass measurement. The default value for this constant temperature setting is 50 °C, which was determined empirically by the manufacturer (Patashnick and Rupprecht, 1991).

In 1997, Allen et al. (1997) reported wide variations in the relationships between the traditional filter-based method and the TEOM. The extent of variations depended upon monitoring location, sampling time, and PM concentrations. Ammonium nitrate (NH₄NO₃) and semivolatile organics (SVOCs) were suspected to be the cause for the observed discrepancies. Immediately following this study, a series of articles, responses, and rebuttals were published to discuss and debate the pros and

cons of the TEOM method for various applications (Allen, 1998; Patashnick, 1998a, 1998b). Since then, more studies reported that the TEOM gave lower PM mass concentration measurements as compared with filter-based PM samplers (Ayers et al., 1999; Charron et al., 2003; Jaques et al., 2004; Lee et al., 2005; Price et al., 2003; Rizzo et al., 2003; Vega et al., 2003). Two plausible reasons discussed extensively from these studies are (1) losses of particulate materials (semivolatile PM and particle-bound water) in the TEOM due to its internal heating mechanism; and (2) negative and positive artifacts associated with changes in air mass composition (Allen et al., 1997) or thermal instability of the TEOM system (Page et al., 2007; Patashnick, 1998a).

To make TEOM measurements comparable with the regulatory PM federal reference method (FRM) samplers, several approaches, such as correction factors (Green, 2001; Green and Fuller, 2006; Green et al., 2001; King et al., 2000; Muir, 2000; Patashnick and Rupprecht, 1991; Rupprecht & Patashnick. Co., 2001), $(\text{NH}_4)_2\text{SO}_4$ loss correction (Charron et al., 2004; Price et al., 2003) instrument settings change (Rupprecht & Patashnick. Co., 2001), and use of the differential TEOM monitor (Jaques et al., 2004; Patashnick et al., 2001), have been studied. Based upon field evaluations at several US and overseas sites (Patashnick and Rupprecht, 1991), the manufacturer recommended the following correlation between TEOM monitors and the reference samplers:

$$Y = 3.0 + 1.03 \times X \quad (1)$$

where Y is the PM_{10} concentration measured by an FRM PM_{10} sampler and X is the PM_{10} concentration measured by a TEOM- PM_{10} monitor with default internal settings ($\mu\text{g}/\text{m}^3$).

To use TEOM PM_{10} data for regulatory compliance with the European Air Quality Standard, Muir refined the correlation factors for different application scenarios (Muir, 2000). However, Muir's study indicated that there was no universal correction factor for different applications. PM sampling location, sampling time and season, PM size, and instrumental

settings all have different significant impacts on the correction factors. Charron et al. (2004) found that both PM_{10} and $\text{PM}_{2.5}$ TEOM data were significantly improved by adding the particulate NH_4NO_3 at a rural site. In 1993, a TEOM temperature setting of 30°C was approved by EPA on a case-by-case basis (Rupprecht & Patashnick. Co., 2001). It was specified that if there is a significant portion of volatile organic compounds (VOCs) in PM_{10} and if the measurements are to be conducted in winter with outdoor temperature less than 25°C , instead of using default setting of 50°C , a lower temperature of 30°C can be used. In 2001, Patashnick et al. (2001) developed a differential TEOM system to overcome the bias of semivolatile PM mass loss. The preliminary data demonstrated its ability to track adsorption and desorption from sample filters (Jaques et al., 2004; Patashnick et al., 2001).

TEOM resonant frequency changes are not only influenced by the collected PM but also by air stream relative humidity (RH) and temperature, gaseous pollutants, and particle characteristics. The absorption of gaseous species and moisture in the sample stream onto the filter will result in positive measurement artifacts. The semivolatile and particle-bound water desorption from the filter will result in negative measurement artifacts. How these positive and negative artifacts are handled has significant impacts on TEOM measurements. The Division of Air Quality at North Carolina Department of Environment and Natural Resources recommended that negative values up to $-10 \mu\text{g}/\text{m}^3$ should be kept and below $-10 \mu\text{g}/\text{m}^3$ should be flagged (2007). But in its standard operating procedure (SOP), no detailed scientific basis was provided for this recommendation. The exact reasons for the positive and negative artifacts in TEOM mass concentration measurements are very complex and not fully understood. Gilbert and Clark (2001) and Jarrett et al. (2001) reported that moisture adsorption and desorption by the filter were the primary sources of negative and positive artifacts. The mechanism for moisture adsorption and desorption can also likewise apply to other semivolatile inorganic and organic compounds. Moisture adsorption and desorption are illustrated in three case scenarios in Figure 1.

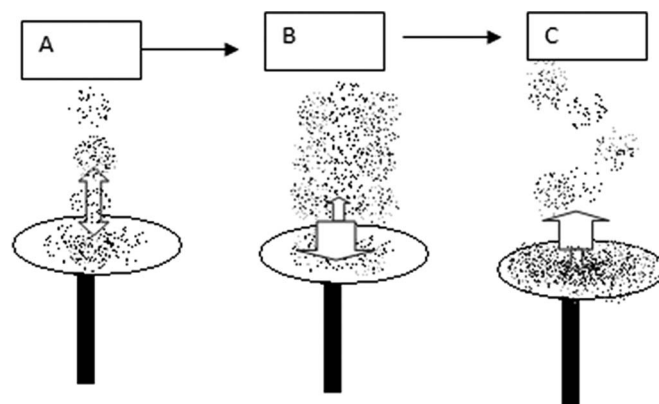


Figure 1. Illustration of three case scenarios to explain positive and negative artifacts from moisture adsorption and desorption in TEOM mass measurement: (A) moisture adsorption and desorption from TEOM filter and collected PM are in equilibrium, the “true” mass concentration is measured; (B) during an increase RH, TEOM filter moisture absorption rate is faster than desorption rate, resulting in a higher TEOM mass concentration measurement ($\Delta M + \text{absorbed moisture}$) than the “true” value (ΔM); (C) during decrease in RH, TEOM filter moisture absorption rate is slower than desorption rate, resulting in a lower mass concentration ($\Delta M - \text{desorbed moisture}$) than the “true” mass (ΔM).

In recent years, animal feeding operations (AFOs) shifted to fewer farms with greater numbers of animals raised on each farm. PM concentration and emission at AFOs have been of increasing concern to the public and the EPA (National Research Council [NRC], 2003). The TEOM has been adapted for indoor PM concentration measurements in some AFO air quality studies (Heber et al., 2006). However, only for PM₁₀ in ambient air, TEOM was approved by US EPA as an equivalent method for 24-hr average concentration (Rupprecht & Patashnick. Co., 2001).

In poultry houses, air consists of a mixture of dry air mass, moisture (H₂O), and several other gases such as carbon dioxide (CO₂), ammonia (NH₃), hydrogen sulfide (H₂S), and VOCs and SVOCs. The gaseous pollutants may present with high concentrations. In-house moisture production includes water vapor evaporated from wet manure and exhaled by the birds. CO₂ originates mainly from air exhaled by the birds, and can be used to assess the effectiveness of ventilation. NH₃ originated mainly from bacteriological process in the manure, and it is easily bound with water to form NH₃·H₂O. In-house NH₃ concentrations depend on ventilation, temperature, RH, bird stocking density, and manure conditions. Particle phase NH₃·H₂O may affect TEOM measurement. VOCs and SVOCs are produced from decomposition of amino acids and carbohydrates (Mackie et al., 1998; Mata-Alvarez et al., 2000). The reported top five VOCs or SVOCs include acetic acid, butanedione, methanol, acetone, and ethanol (Trabuea et al., 2010). High PM concentrations were observed in poultry houses, and the influencing factors on PM concentrations in poultry houses include ventilation, bird activity, bird age and type, air temperature, RH, manure conditions, feeding system, hygiene, and manure management practices (Li et al., 2011). Using TEOM for PM measurements in poultry houses faces some harsh environments, which the instrument was not originally designed to withstand.

For data quality assurance, there is an urgent need to investigate possible uncertainties, and limitations associated with TEOM measurements in AFO environments so that the measurements by TEOM and FRM methods can be comparable with each other. The objectives of this study were to (1) compare PM measurements by TEOMs and gravimetric filter-based PM samplers; (2) investigate the influence of TEOM internal temperature settings on PM measurements; and (3) assess the effects of instrumental and environmental variables on PM measurements in AFO-poultry environments.

Methodology

PM Monitors and Samplers: TEOM-PM_{2.5}/PM₁₀/TSP Monitors, Partisol 2300 PM_{2.5} Speciation Sampler, Low-Volume PM₁₀ Sampler

The TEOM (1400A; Thermo Scientific, Franklin, MA, USA) is a “gravimetric” instrument that measures PM mass concentration using a tapered element oscillating microbalance. It draws air through a filter, continuously weighs the filter, and calculates mass concentrations. The fundamental measurement principle by the tapered oscillating element is based upon the following equation (Rupprecht & Patashnick. Co., 2001):

$$\Delta M = K_0 \left[\frac{1}{f_t^2} - \frac{1}{f_0^2} \right] \quad (2)$$

where M is the mass change in g, f_0 is measured initial frequency in Hz, f_t is measured frequency at time t in Hz, and K_0 is the characteristic constant of each instrument. In this study, for PM_{2.5}, PM₁₀, or total suspended particle (TSP) measurements, the TEOM was operated with PM_{2.5}, PM₁₀, or TSP sampling inlets, respectively.

The Partisol 2300 PM_{2.5} speciation sampler (Thermo Scientific, Franklin, MA) is one type of filter-based PM_{2.5} sampler. This speciation sampler consists of four filter sample cartridges and a flow control system. These cartridges can hold quartz filters for organic carbon (OC) and element carbon (EC) analysis, nylon filters for ion (cation and anion) analysis, and Teflon filters for trace element and PM_{2.5} mass analysis, respectively. Each cartridge contained a sharp-cut PM_{2.5} impactor operating at a flow rate of 10.0 L/min (for nylon and quartz filters) or 16.7 L/min (for Teflon filters). In this study, 24-hr PM_{2.5} samples collected on the Teflon filters were analyzed for mass concentrations following EPA's standard operating procedures for the National Chemical Speciation Network (CSN) at RTI International (RTI International, 2003). The gravimetric analysis of PM_{2.5} samples was conducted in a temperature- and RH-controlled chamber. The temperature was controlled between 20 and 23 °C with a standard deviation less than 2 °C. The RH was controlled between 30% and 40% with a standard deviation less than 5%. The filters (46.2 mm diameter; Whatman 2-μm Polytetrafluoroethylene [PTFE]; Whatma Inc., Clifton, NJ) were conditioned in this chamber for a minimum of 48 hr prior to determining the pre- and post-weights of the filter using a microbalance with a minimum readability of ±1 μg and a repeatability of 1 μg. National Institute of Standards and Technology (NIST)-traceable standards were applied to determine the microbalance performance.

The filter-based low-volume PM₁₀ sampler (LV-PM₁₀) consisted of a FRM PM₁₀ sampling head and flow control system (Wang et al., 2005). The LV-PM₁₀ flow control system was designed and manufactured by Texas A&M University (Wang et al., 2005). The flow rate through the PM₁₀ inlet was 1 m³/hr. This flow rate was maintained with less than 1% variation by monitoring pressure drop across a sharp-edge orifice meter. The PM₁₀ sampling duration for the 46.2 mm diameter 2-μm PTFE was 24 hr. The total volume of air was calculated based on sampling time and air flow rate (Wang et al., 2005). The sample filters were kept in plastic Petri dishes and conditioned before and after testing in a weighing chamber for 24 hr at 20–23 °C and 30–40% RH for pre- and post-weights. Each filter sample was weighed three times, and the mean was used to calculate PM₁₀ concentrations.

PM Samplers Placement and Settings

The field PM sampling and monitoring were conducted in a high-rise layer house at a commercial egg production farm in North Carolina. It was a typical tunnel-ventilated house in the Southeast United States with dimensions of 175 m (length) by 18 m (width). The house contained approximately 95,000 hens in six rows of 4-tier A-frame and curtain-backed cages on the upper floor. Manure fell onto the curtain-backed cages and then

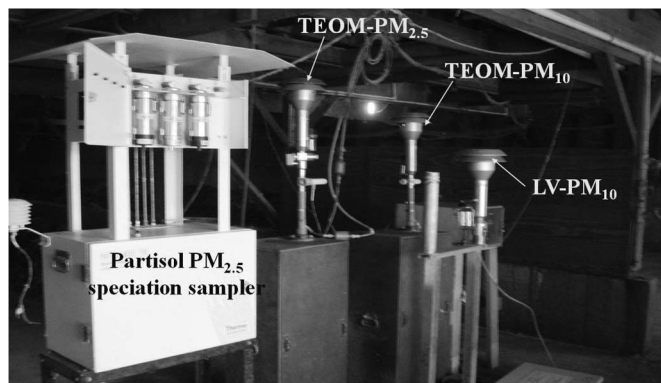


Figure 2. Collocated PM samplers on the first floor of the high-rise layer house (from left to right): Partisol PM_{2.5} speciation sampler, TEOM PM_{2.5}, TEOM PM₁₀, and LV-PM₁₀.

down into the first floor (pit), where it was stored for approximately one year.

As illustrated in Figure 2, for comparison of TEOM and filter-based samplers, the TEOM-PM₁₀, TEOM-PM_{2.5}, LV-PM₁₀, and Partisol 2300 PM_{2.5} speciation samplers were placed immediately upstream of the primary representative exhaust fan on the first floor of the layer house. The sampler inlets were 2 m away from the fans, where airflow generated by the exhaust fans was observed to be low (less than 2 m/sec) and had insignificant impact on the samplers' performance. In this set of comparison studies, the internal temperatures of two TEOMs were set at 50 °C (the manufacturer default value). Fifteen filter-based 24-hr samples were taken for PM₁₀, and 13 filter-based 24-hr samples were taken for PM_{2.5} during November 2009 to December 2009.

To investigate the effect of TEOM operating temperature (internal temperature setting), the two side-by-side TEOMs (Figure 2) were tested with both units equipped with TSP, PM₁₀, or PM_{2.5} sampling heads. One TEOM (labeled as TEOM 1) was set at the default temperature setting (50 °C). The other TEOM (labeled as TEOM 2) was set at either 30 °C or no internal heating. No internal heating means that the internal temperature control of this TEOM was turned off (in inactive mode). The continuous data collection occurred between October 2009 and December 2009.

Measurements of Environmental Parameters

In addition to PM mass concentration measurements, some environmental parameters were also measured to investigate their potential impacts on TEOM measurements. Surrounding air temperature and RH were measured by capacitance-type RH/T probes located in the house beside the TEOM units. Concentrations of NH₃ and CO₂ at this location were also measured with a photoacoustic infrared multigas monitor (INNOVA Model 1412; LumaSense Technologies A/S, Ballerup, Denmark). From December 2008 to December 2009, 66 PM_{2.5} nylon filter samples were taken using the Partisol 2300 speciation sampler for analysis of anion (SO₄²⁻, Cl⁻, and NO₃⁻) and cations (NH₄⁺, Na⁺, and K⁺). The ion analysis was conducted using ion chromatography (IC) by RTI International at Research Triangle Park (RTP), North Carolina.

TEOM Operation and Data Validation

The negative and positive artifacts/biases can be minimized when the average concentrations are calculated over longer periods of instantaneous monitoring data. The longer periods can smooth out short-term fluctuations and highlight longer-term trends. Meanwhile, the effects of moisture and volatile compounds could be minimized by limiting filter loading because accumulated PM results in more adsorption (Heber et al., 2006). In this study, the TEOM filters were replaced when the loading reached 50%, and 30-min average data were recorded. Except for some ghost peaks, both positive and negative artifacts were kept in average mass concentration calculations. Deleting of negative values results in overestimating mass concentration and therefore should be avoided.

Statistical Analysis

All univariate data analyses, *t* tests, and linear models for this study were conducted using SAS/STAT software, version 9.2 (SAS Institute Inc., Cary, NC, USA).

Results and Discussion

TEOM versus Gravimetric Methods

Table 1 presents the means and medians of PM₁₀ and PM_{2.5} mass concentrations from the filter-based samplers (LV and Partisol) and the TEOMs. During sampling period, the temperature ranged from 18.0 to 22.0 °C, with mean 20.2 °C and standard deviation 1.2 °C, and the relative humidity (RH) ranged from 56% to 79%, with mean 70% and standard deviation 5.6%. The TEOM reported lower PM₁₀ and PM_{2.5} mass concentrations than the filter-based gravimetric method. Also robust estimators (medians) were calculated (Table 1), and they were influenced less by skewed data (occasional low/high-concentration data) than the TEOM means. The TEOM and LV sampler had similar median values of PM₁₀, but significantly different medians of PM_{2.5}.

Both paired *t* test and nonparametric sign and signed rank tests were conducted to test the difference between these two methods (SAS, 2009). The paired *t* test looks for a difference in means and assumes the observations are independent and identically normally distributed. Although the nonparametric signed and signed-rank tests look for a difference in medians, neither of these tests assume the data to be normally distributed. For PM₁₀ comparisons, the *P* values for the paired *t* (0.029), signed (0.035), and signed-rank (0.021) tests were less than 0.05 (Table 1). Thus, there are significant differences between the PM₁₀ means and medians of two methods. This observation agrees with other studies (Allen et al., 1997; Park et al., 2006; Rizzo et al., 2003; Vega et al., 2003). Also, for PM_{2.5}, statistical tests show the two methods had significantly different means (*P* < 0.05).

Reasonable strong correlation between PM₁₀ concentrations determined by the TEOM and the LV sampler were obtained (Figure 3A), and is expressed by eq 3.

Table 1. PM₁₀ and PM_{2.5} concentration comparison

	PM ₁₀ Concentration (μg/m ³ , <i>n</i> * = 15)	
	TEOM-PM ₁₀	LV-PM ₁₀
Mean ± SD	902 ± 263	968 ± 284
Median	974	977
Test for difference between methods of TEOM-PM ₁₀ and LV-PM ₁₀ samplers		
Test	Statistic	<i>P</i> Value
Student's <i>t</i> (<i>t</i>)	2.4	0.029
Sign (<i>M</i>)	4.5	0.035
Signed rank (<i>S</i>)	40	0.021
	PM _{2.5} Concentration (μg/m ³ , <i>n</i> * = 13)	
	TEOM-PM _{2.5}	Speciation (Partisol) PM _{2.5}
Mean ± SD	125 ± 41.0	185 ± 64.0
Median	132	192
Test for difference between: TEOM-PM _{2.5} and Partisol PM _{2.5}		
Test	Statistic	<i>P</i> Value
Student's <i>t</i> (<i>t</i>)	4.89	0.0004
Sign (<i>M</i>)	5.50	0.0034
Signed rank (<i>S</i>)	43.5	0.0007

Note: **n* = number of samples.

$$PM_{10TEOM} = 0.858PM_{10LV} + 71.3, R^2 = 0.86 \quad (3)$$

where PM_{10TEOM} is the PM₁₀ mass concentrations measured by the TEOM μg/m³, and PM_{10LV} is the PM₁₀ mass concentrations measured by the LV-PM₁₀ sampler μg/m³.

The linear fit between PM_{2.5} concentrations measured by the TEOM and Partisol samplers is shown in Figure 3B. PM_{2.5} data were more scattered than PM₁₀, and the linear regression between those two methods is given in eq 4:

$$PM_{2.5TEOM} = 0.443PM_{2.5Par} + 43.8, R^2 = 0.48 \quad (4)$$

where PM_{2.5TEOM} is the PM_{2.5} mass concentrations measured by the TEOM μg/m³, and PM_{2.5Par} is the PM_{2.5} mass concentrations measured by the Partisol PM_{2.5} sampler μg/m³.

Intercepts close to 0 and slopes close to 1 indicate equivalency of the methods. In the two regression models (eqs 3 and 4), large intercepts (71.3 for PM₁₀ and 43.8 for PM_{2.5}) suggest a systematic bias between these two methods. In this study, the PM₁₀ ratio (TEOM/gravimetric) averaged 0.936 (SD = 0.112), and varied between 0.704 and 1.148. The PM_{2.5} ratio (TEOM/gravimetric) averaged 0.738 (SD = 0.319) and varied between 0.373 and 1.672. The variability of the ratios further indicates the lack of agreement between TEOM and gravimetric methods.

The most frequently identified cause of difference in PM mass concentration measurements by filter-based and TEOM methods is the loss of semivolatile mass and PM-bound moisture due to heating of the TEOM sampling stream to 50 °C (Allen et al., 1997; Grover et al., 2005; Jerez et al., 2006; Vega et al., 2003). Therefore, to correct for possible loss of particulate NH₄NO₃ by heating, the concentration of particulate NH₄NO₃ was used to adjust TEOM measurements. The mean concentration of particulate NH₄NO₃ in the testing layer house was observed to be 0.843 μg/m³. Adding NH₄NO₃ to the TEOM-PM_{2.5} concentration did not make any significant improvement to the comparison between TEOM and filter-based methods. Compared with the PM_{2.5} mass concentration (mean = 195 μg/m³), the particulate NH₄NO₃ concentration (mean = 0.843 μg/m³) in the layer house was negligible. NH₄NO₃ was not the dominant compound, and did not contribute significantly to the overall PM mass. Therefore, a substantial portion of mass loss may have been from the volatilization of PM-bound moisture and VOCs/SVOCs. The National Air Emissions Monitor Study (NAEMS) at this research farm detected 20 most prevalent VOCs in canister samples taken at the same sampling location as it for this study. The top five VOCs were acetaldehyde, pentane, 2-butanone, isopropyl alcohol, and dimethyl sulfide (Wang et al., 2010). Total VOC concentration ranged from 0.26 to 0.81 mg/m³ (Wang et al., 2010).

The results of this field study show that TEOM and filter-based methods are not interchangeable in determining PM concentrations in layer houses. Also, this field comparison results motivated a follow-up study on the effects of TEOM internal temperature settings on PM measurements.

Effects of Internal TEOM Temperature Settings

Lower PM with the TEOM as compared with filter-based sampling was hypothesized to be due to heating of the TEOM sampling stream (usually 50 °C). The PM lost in the sampling stream of the TEOM was assumed to be mainly SVOCs and PM-bound water. However, the amount of adsorption or desorption cannot be directly quantified. Temperature-dependent studies of the dynamics of adsorption and desorption may provide some insight on the impacts of adsorption or desorption. Therefore, two TEOM units were tested side by side (with internal temperature settings of 50 and 30 °C), and the mass concentrations measured by the units were compared.

Prior to the comparison test, two identical TEOM monitors (Model 1400A) were tested side by side for TSP and PM₁₀ for a total 4 weeks to assess agreement.

Figures 4 and 5 show one example for each of TSP and PM₁₀ collocation test results. The TEOM 1 and TEOM 2 had very similar mass concentration time series (Figures 4A and 5A), Figure 4B and 5B showing agreement of the two units. The TEOM 1 and TEOM 2 were highly correlated (*R*² > 0.9). Also the paired *t* test showed there were no significant differences between the TEOMs for TSP and PM₁₀ (TSP: *P* = 0.139 and 95% confidence interval [CI] = [−5.39, 38.4]; PM₁₀: *P* = 0.08 and 95% CI = [−16.8, 0.96]). Overall, the repeatability of the two instruments was very good, and both tracked well over the entire testing period.

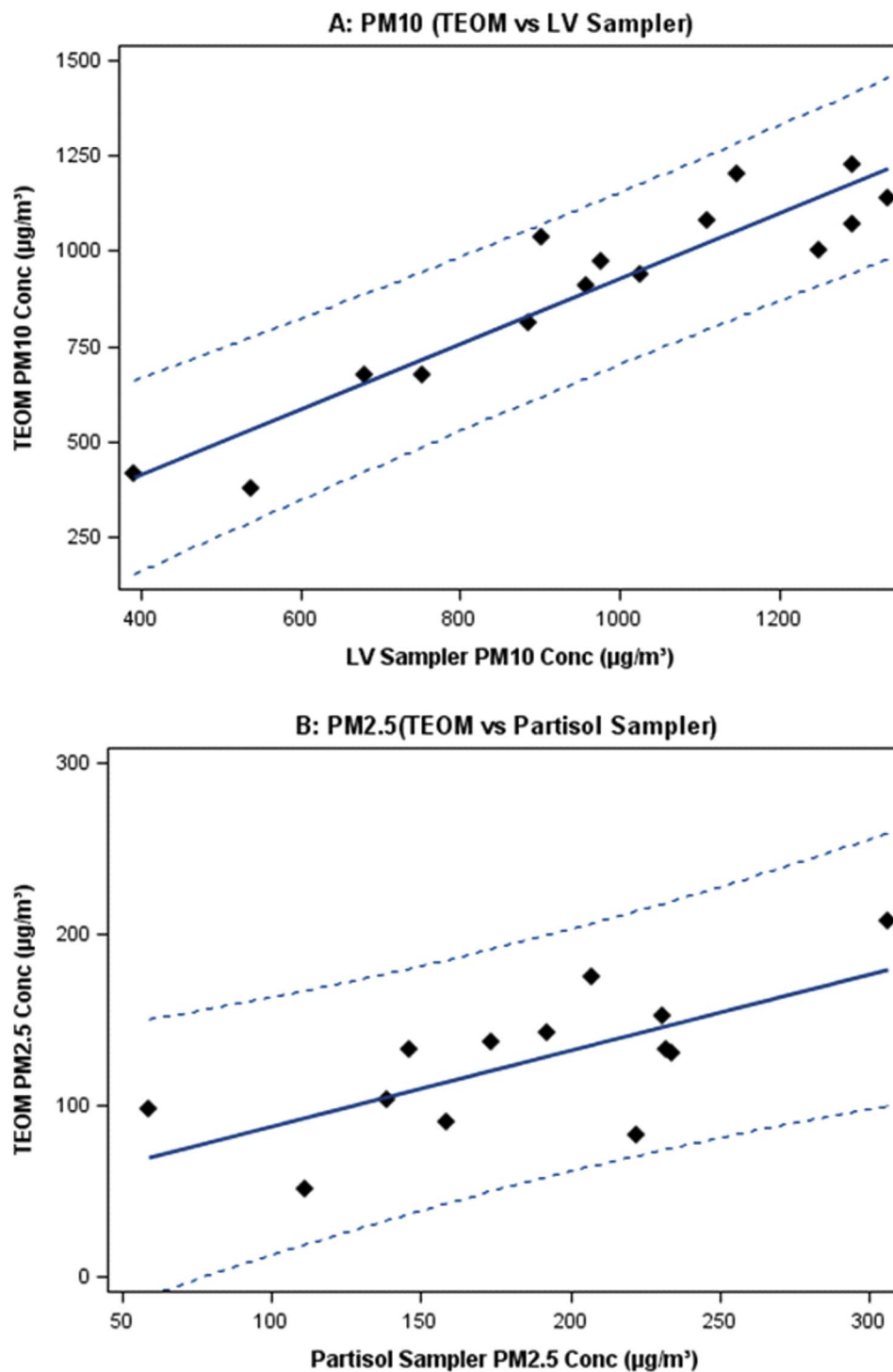


Figure 3. Comparisons of PM_{10} and $\text{PM}_{2.5}$ measurements by the TEOMs and the filter-based samplers. Solid line = regression fit and dashed-line = 95% prediction limits.

Figure 6 shows the comparison of $\text{PM}_{2.5}$, PM_{10} , and TSP measured by TEOM 1 at 50 °C and TEOM 2 at 30 °C. It is observed from Figure 6A that the TEOM operating at 50 °C measured much lower mass $\text{PM}_{2.5}$ concentrations as compared with 30 °C. For PM_{10} (Figure 6B) and TSP (Figure 6C), the magnitudes of the differences in mass concentrations from the TEOMs operated at 50 and 30 °C were smaller than for $\text{PM}_{2.5}$. This lower magnitude of the differences in PM_{10} or TSP than in

$\text{PM}_{2.5}$ is consistent with the losses of SVOCs and bound moisture associated with the fine PM, because SVOCs and PM-bound moisture have larger proportions in the fine PM fraction than in larger particles (Finlayson-Pitts and Pitts, 1999). Statistical analyses (Table 2) showed significant differences between these two TEOMs for $\text{PM}_{2.5}$ measurements ($P < 0.0001$). The mean difference was $59.3 \mu\text{g}/\text{m}^3$ (50 vs. 30 °C) and the relative difference percentage (RDiff) was 49.2%, which means almost half of $\text{PM}_{2.5}$

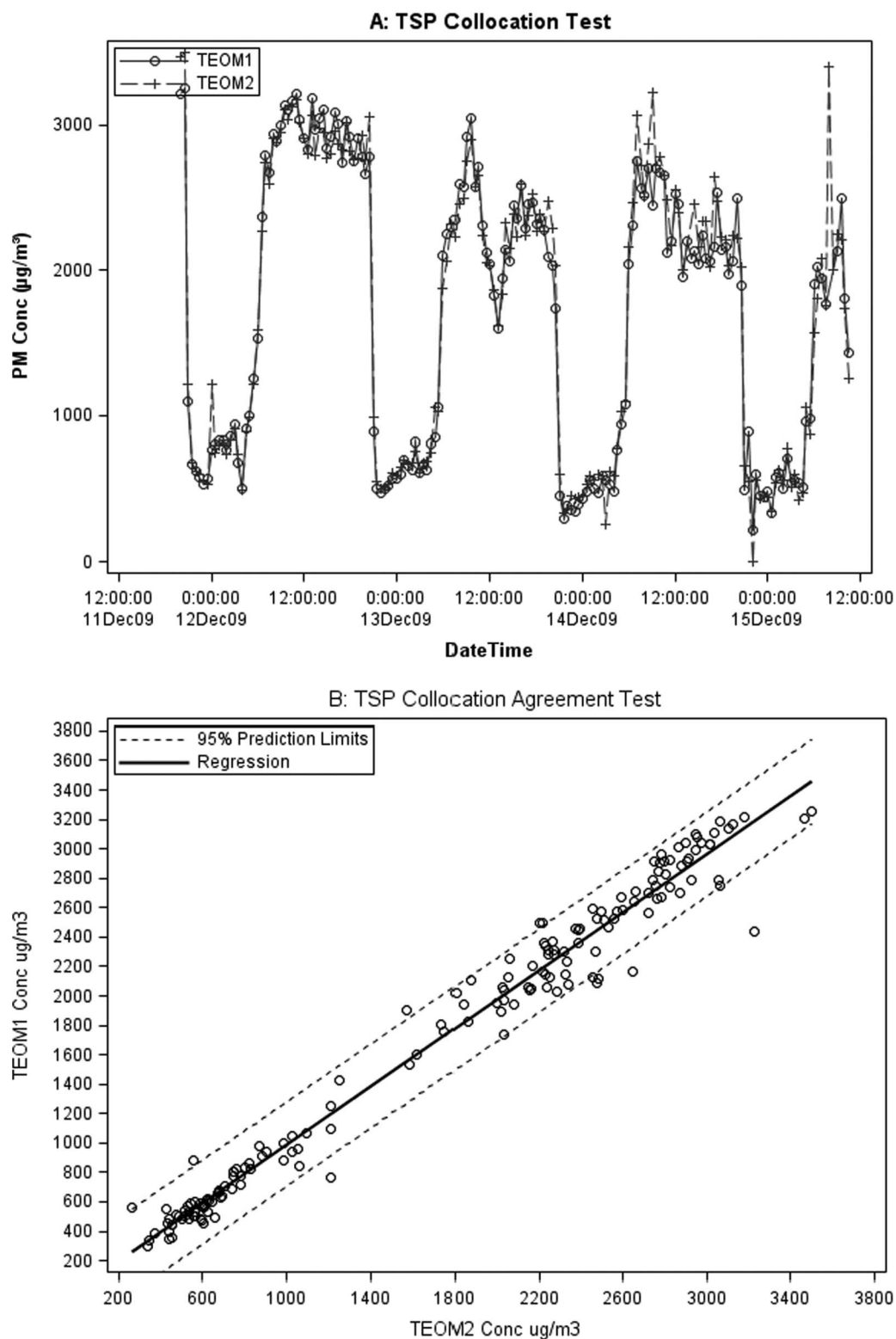


Figure 4. TEOM collocation test results for TSP: (A) TSP collocation test time profile for TEOM 1 and TEOM 2 and (B) the agreement test of TEOM 1 and TEOM 2 (solid line = regression line and dashed lines = 95% prediction limits).

mass was lost when TEOM operation temperature setting changed from 30 to 50 °C. For PM_{10} measurements, the mean difference was $58.6 \mu\text{g}/\text{m}^3$ and RDiff was 5.72%. For TSP, the mean difference was $182 \mu\text{g}/\text{m}^3$ and RDiff was 11.0%. Higher errors in

TSP than PM_{10} was possibly due to higher frequency noise error (TSP noise level of 0.46 vs. PM_{10} noise level of 0.16) and larger standard errors. The noise is an indicator of how well the mass transducer is performing, and it is considered to be normal if less

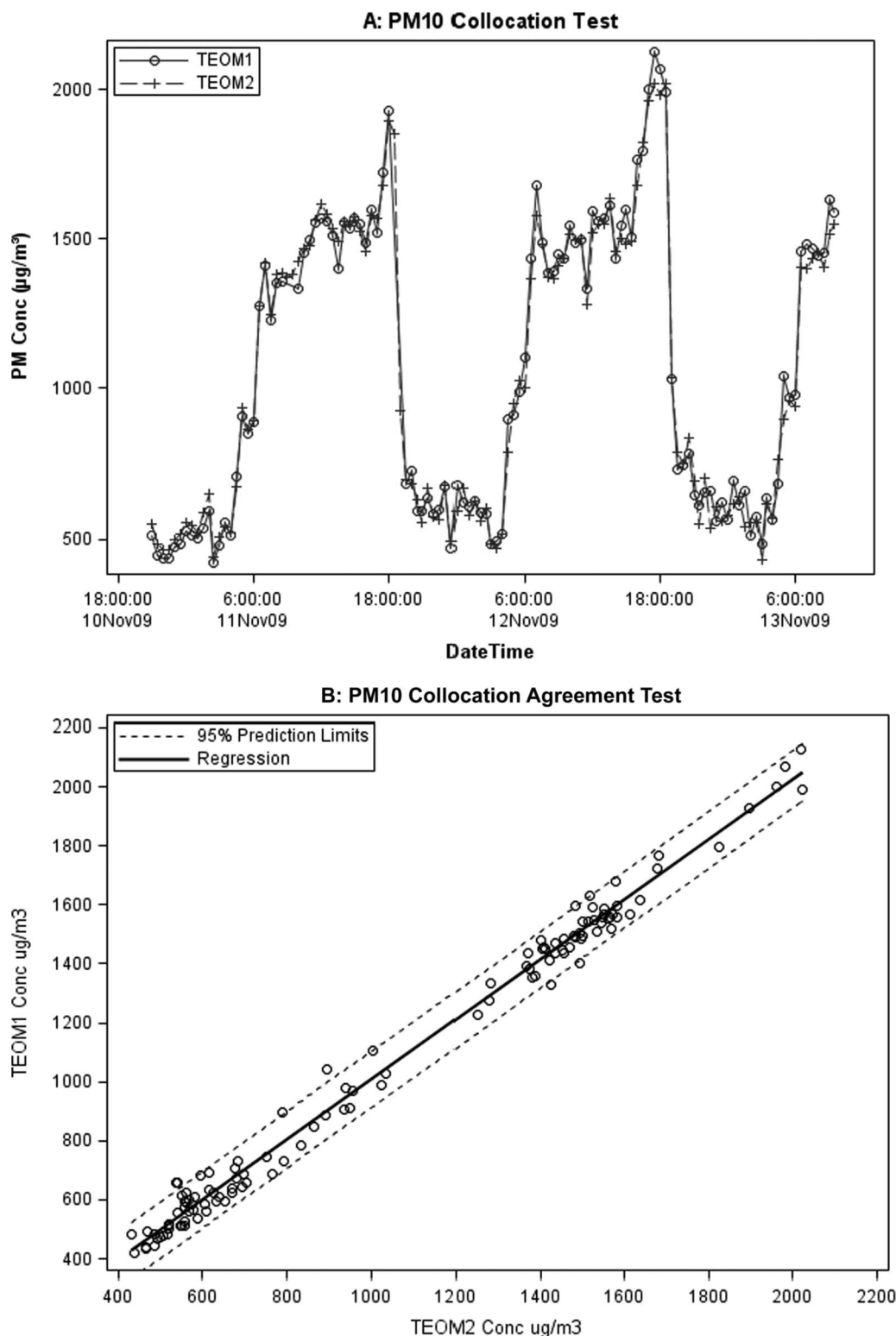


Figure 5. TEOM collocation test results for PM₁₀: (A) PM₁₀ collocation test time profile for TEOM 1 and TEOM 2 and (B) the agreement test of TEOM 1 and TEOM 2 (solid line = regression line and dashed lines = 95% prediction limits).

than 0.10. The results show that TEOM internal temperature settings have significant impact on PM mass concentrations.

Figure 7 shows the comparison of PM_{2.5}, PM₁₀, and TSP concentrations measured by TEOM 1 (50 °C) and TEOM 2

(temperature control was inactive). It was observed that in-house temperature variation had big effect on the resonant frequency of the tapered element. This is apparently the reason for the negative and positive artifacts in the TEOM signals (Figure 7). For TEOM 2,

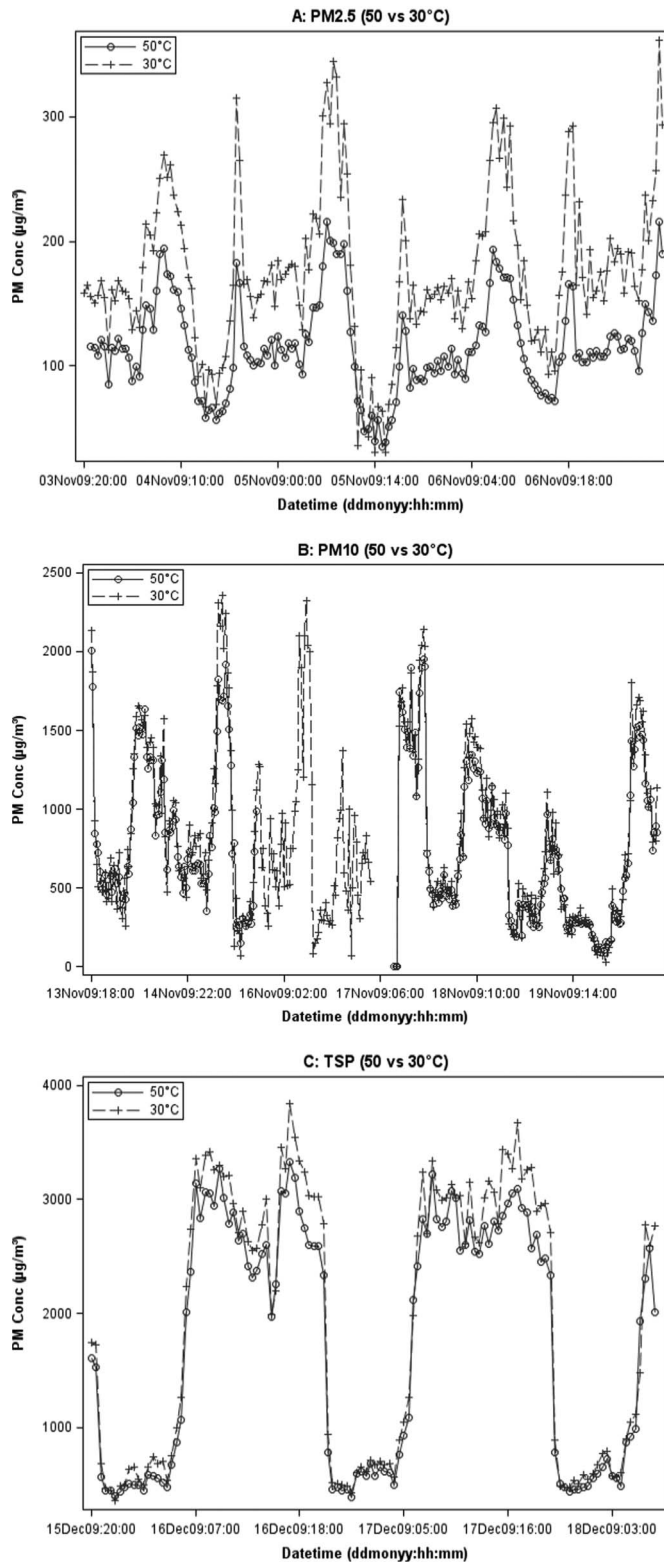


Figure 6. The effects of internal temperature setting on TEOM measurements: (A) PM_{2.5} 50 °C versus 30 °C test; (B) PM₁₀ 50 °C versus 30 °C test; and (C) TSP 50 °C versus 30 °C test.

there were 16% negative mass signal for PM_{2.5}, 7.8% for PM₁₀, and 36.1% for TSP. When TEOM temperature control is inactive, air mass composition changes and thermal instability cause negative/positive artifacts (Allen, 1998; Allen et al., 1997; Patashnick,

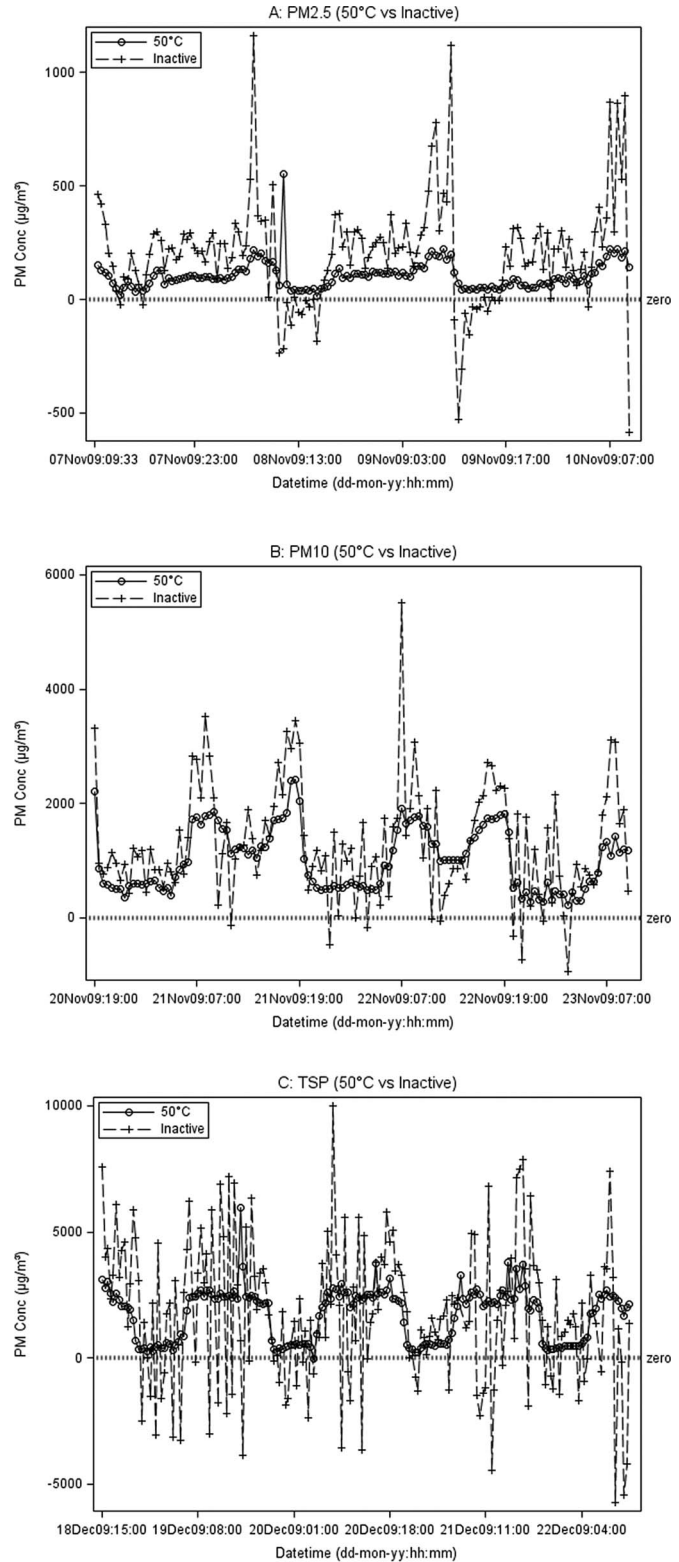


Figure 7. TEOM temperature settings tests: 50 °C versus inactive; (A) PM_{2.5} (B) PM₁₀, and (C) TSP.

1998a, 1998b). Also as shown in Figure 7, both positive and negative amplitudes of the response increased when TEOM's temperature control was inactive. Moreover, when the temperature control was inactive, there were higher noise levels. When TEOM

Table 2. Effects of TEOM internal temperature settings (30 vs. 50 °C) on PM measurements and the statistical test results

	Diff Conc = TEOM 2 ^a – TEOM 1 ^a		
	Diff Conc ($\mu\text{g}/\text{m}^3$)		RDiff ^b (%)
	Mean $\pm$ SD	P Value	
PM _{2.5} ($n^c = 166$)	59.3 $\pm$ 29.6	<0.0001	49.2 $\pm$ 18.3
PM ₁₀ ($n^c = 250$)	58.6 $\pm$ 122	<0.0001	5.72 $\pm$ 17.5
TSP ($n^c = 120$)	182 $\pm$ 186	<0.0001	11.0 $\pm$ 8.71

Notes: ^aTEOM 1: temperature setting = 50 °C; TEOM 2: temperature setting = 30 °C. ^bRDiff = Diff Conc/TEOM 1 $\times$ 100 = (TEOM 2 – TEOM 1)/TEOM 1 $\times$ 100. ^cNumber of measurements (30-min mass concentration averages).

Table 3. Effects of TEOM internal temperature settings (inactive vs. 50 °C) on PM measurements and the statistical test results

	Diff Conc = TEOM 2 ^a – TEOM 1 ^a (With Negative Reading)		
	Diff Conc ($\mu\text{g}/\text{m}^3$)		RDiff ^b (%)
	Mean $\pm$ SD	P Value	
PM _{2.5} ($n^c = 144$)	100 $\pm$ 218	<0.0001	55.9 $\pm$ 240
PM ₁₀ ($n^c = 125$)	271 $\pm$ 646	<0.0001	27.0 $\pm$ 102
TSP ($n^c = 120$)	–552 $\pm$ 1.971 $\times 10^3$	<0.0001	–21.6 $\pm$ 330

Notes: ^aTEOM 1: temperature setting = 50 °C; TEOM 2: temperature setting = inactive. ^bRDiff = Diff Conc/TEOM 1 $\times$ 100 = (TEOM 2 – TEOM 1)/TEOM 1 $\times$ 100. ^cNumber of measurements (30-min mass concentration averages).

operated at 50 °C, the noise values were normally in the range of 0.1 to 2.0 (higher than ambient measurement noise level). However, when TEOM operated at ambient conditions (temperature control was inactive), the noise values varied from 0.5 to 20. Such high noise level interfered with the signal.

Statistical analysis (Table 3) showed there were significant differences between the TEOMs for PM_{2.5} measurements ($P < 0.0001$). The mean difference was 100 $\mu\text{g}/\text{m}^3$ (50 °C vs. inactive) and the relative difference percentage (RDiff) was 55.9%. For PM₁₀ measurements, the mean difference was 271 $\mu\text{g}/\text{m}^3$ and RDiff was 27.0%. For TSP, the mean difference was –552 $\mu\text{g}/\text{m}^3$ and RDiff was –21.6%. The significant disagreement of TEOM 1 and TEOM 2 for TSP may have been due to high noise level from TEOM 2.

The TEOM measurement is based upon the change in the tapered element's natural resonance frequency and spring constant of K_0 . The “Young's modulus” (a measure of the stiffness) of the tapered element changes with the temperature. Therefore, the spring constant of the tapered element (K_0) will change as the temperature changes, thus induces thermal bias for TEOM measurement. The

inactive mode reflects environmental temperatures, and therefore will be more subject to the surrounding air temperature fluctuation effects. The inactive model was used in this study for comparison purpose only. It is not recommended in the actual operations. Maintaining a constant sample temperature is critical. In the future, some advanced computer algorithms can be applied to analyze and transform the signal while suppressing the effects of noise, so useful information contained in the raw signal can be obtained. These experimental results indicate that the TEOM monitor could not operate correctly when the internal heating was turned off.

Effects of Environmental and Instrumental Parameters

The preceding results motivated a further analysis of the data to assess the effects of environmental and instrumental factors (called predictor variables) on TEOM measurements (called the response). The environmental factors included RH, in-house temperature (Temp), and NH₃ and CO₂ concentrations. NH₃ and CO₂ were used in this study as practical substitutes to assess the VOC effect on PM measurements. VOCs are produced from the degradation of amino acids and carbohydrates in birds and from incomplete anaerobic digestion of manure (Mackie et al., 1998; Mata-Alvarez et al., 2000). It is very difficult to quantify VOCs accurately in AFOs at present. On the other hand, CO₂ are produced mainly from air exhaled by the birds, and NH₃ are produced mainly from bacteriological process in the manure. The NH₃ and CO₂ may reflect the variation of VOC emission intensity in AFOs. The instrumental factors include (1) the difference of noise, which is an indicator of how well the mass transducer is performing; and (2) the difference in filter loading, which is an indicator of the filter's total capacity. The descriptive statistics of the environmental and instrumental factors are listed in Table 4. The PM₁₀ and TSP instrument noises were higher than the defined normal value of 0.1. Also, the layer house had high levels of RH (61% in the layer house vs. 50% annual average ambient RH in North Carolina), NH₃ (60 ppm in the layer house vs. several ppb in the background ambient locations), and CO₂ (2000 ppm in the poultry house vs. 300–400 ppm in the background ambient locations).

The relationship between the environmental and instrumental factors on the difference of PM mass concentrations between two TEOMs is shown in Figure 8. From this figure, it appears that there are some relationships between the predictor variables and the response. By using regression analysis (Johnson and Wichern, 2008), the effects of the predictor variables on the response were estimated (Table 5) for PM_{2.5}, PM₁₀, and TSP. Variance inflation factors (VIFs) were used to diagnose independent variable multicollinearity because multicollinearity of the independent variables provides redundant information and it affects the parameter estimates (Tao and Walsh, 2005). The VIFs (Table 5) are not quite large for all six independent variables, and they do not indicate the presence of strong collinearity. The estimate P values associated with the slopes for six independent variables indicate (Table 5) (a) for PM_{2.5}, the slopes of noise diff, RH, and CO₂ were significantly different from zero at $\alpha = 0.1$; (b) for PM₁₀, the slopes of noise diff, and RH were

Table 4. Descriptive statistics of the environmental and instrumental factors

	PM _{2.5} (<i>n</i> ^a = 78)		PM ₁₀ (<i>n</i> ^a = 120)		TSP (<i>n</i> ^a = 56)	
	Mean ± SD ^b	Min/max ^c	Mean ± SD ^b	Min/max ^c	Mean ± SD ^b	Min/max ^c
T2_filter loading ^d	23.7 ± 1.71	21.0/27.0	34.5 ± 13.6	18.5/64.0	22.5 ± 2.76	19.0/29.0
T1_filter loading ^e	21.2 ± 1.57	18.0/24.0	43.2 ± 24.3	18.0/87.0	21.6 ± 3.05	18.0/27.5
Diff filter loading ^f	2.57 ± 0.48	2.00/3.00	−8.23 ± 11.1	−31.0/1.00	0.89 ± 0.59	−0.50/2.00
T2_noise ^g	0.05 ± 0.02	0.02/0.11	0.16 ± 0.12	0.03/0.83	0.46 ± 0.27	0.08/1.25
T1_noise ^h	0.03 ± 0.01	0.02/0.06	0.12 ± 0.08	0.03/0.62	0.30 ± 0.15	0.07/0.73
Diff Noise ⁱ	0.01 ± 0.02	−0.02/0.05	0.04 ± 0.08	0.03/0.62	0.16 ± 0.19	−0.14/0.77
RH ^j (%)	61.1 ± 7.12	44.0/70.7	74.6 ± 4.95	63.1/85.5	71.6 ± 7.12	55.7/79.2
Temp ^k (°C)	18.9 ± 1.51	15.8/21.6	21.4 ± 0.83	19.8/23.1	18.3 ± 1.51	15.9/21.2
NH ₃ (ppm)	59.6 ± 27.2	5.55/130	65.1 ± 31.4	10.8/145	52.8 ± 25.7	9.69/99.6
CO ₂ (ppm)	2446 ± 557	949/3569	2363 ± 734	1071/4748	2720 ± 432	998/3568

Notes: ^aData points; ^bstandard deviation; ^cmaximum and minimum values; ^dTEOM 2 (50 °C) filter loading; ^eTEOM 1 (30 °C) filter loading; ^fthe filter loading difference between TEOM 1 and TEOM 2; ^gTEOM 2 (50 °C) instrument noise level; ^hTEOM 1 (30 °C) instrument noise level; ⁱthe noise difference between TEOM 1 and TEOM 2; ^jrelative humidity in house; ^ktemperature in house.

significantly different from zero at $\alpha = 0.1$; and (c) for TSP, only the slope of CO₂ was significantly different from zero at $\alpha = 0.1$.

Based upon Least absolute shrinkage and selection operator (LASSO) method of Tibshirani (1996), the best general linear models are

$$\text{Diff Conc} = -49.0 + 276 \text{ Diff Noise} + 1.27 \text{ RH} + 0.01 \text{ CO}_2, \quad R^2 = 0.35 \quad (5)$$

PM₁₀:

$$\text{Diff Conc} = 289 - 0.39 \text{ Diff filter loading} - 253 \text{ Diff Noise} - 3.10 \text{ RH}, \quad R^2 = 0.13 \quad (6)$$

TSP:

$$\text{Diff Conc} = 566 - 2.57 \text{ RH} - 0.07 \text{ CO}_2, \quad R^2 = 0.13 \quad (7)$$

where Diff Conc is the difference of mass concentrations measured by two TEOMs (concentrations at 30 °C minus concentrations at 50 °C), Diff Noise is the difference of noises from two TEOMs (noise at 30 °C minus noise at 50 °C), Diff filter loading is the difference of filter loading from two TEOM, RH is in-house relative humidity, and CO₂ is in-house carbon dioxides concentrations in ppm.

The percentages of the variability in mass difference (Diff Conc) explained by these models are approximately 35%, 13%, and 13% for PM_{2.5}, PM₁₀, and TSP, respectively. It is obvious that these environmental and instrumental factors did not explain well the mass differences. Perhaps some unknown important factors existed but were not included in the models. In addition to these additive models (eqs 5–7), models with interactions between variables were also tested, and the interactions among these variables were not significant.

Although there was some evidence showing impacts of environmental and instrumental parameters on TEOM measurement (especially for PM_{2.5}), it is very difficult to quantify and to

minimize their impacts. How to reduce errors and increase the accuracy remains a great challenge for TEOM and FRM data comparison.

Conclusions

In this study, a field evaluation of TEOM PM measurements in a high-rise layer house was conducted to investigate possible uncertainties and limitations associated with TEOM measurements in poultry operation environment. It was observed that TEOM gave lower readings of both PM₁₀ and PM_{2.5} mass concentrations as compared with measurements by the filter-based gravimetric method in a poultry house. The inclusion of particulate NH₄NO₃ did not significantly improve the comparison of TEOM-PM_{2.5} measurement and filter-based PM_{2.5} measurement. The most likely explanations for the difference between these two methods are the TEOM internal heating, moisture, and VOC adsorption and desorption.

For TSP, PM₁₀, and PM_{2.5} measurements, TEOM internal temperature setting has significant effect. Higher PM mass concentrations were measured at lower instrumental temperature setting. The influences of environmental and instrumental parameters were also assessed, and their significances varied case by case. The impacts of the environmental and instrumental parameters on TEOM measurement were very difficult to quantify. For AFO-related air quality study, making TEOM and FRM measurements comparable remains to be a great challenge.

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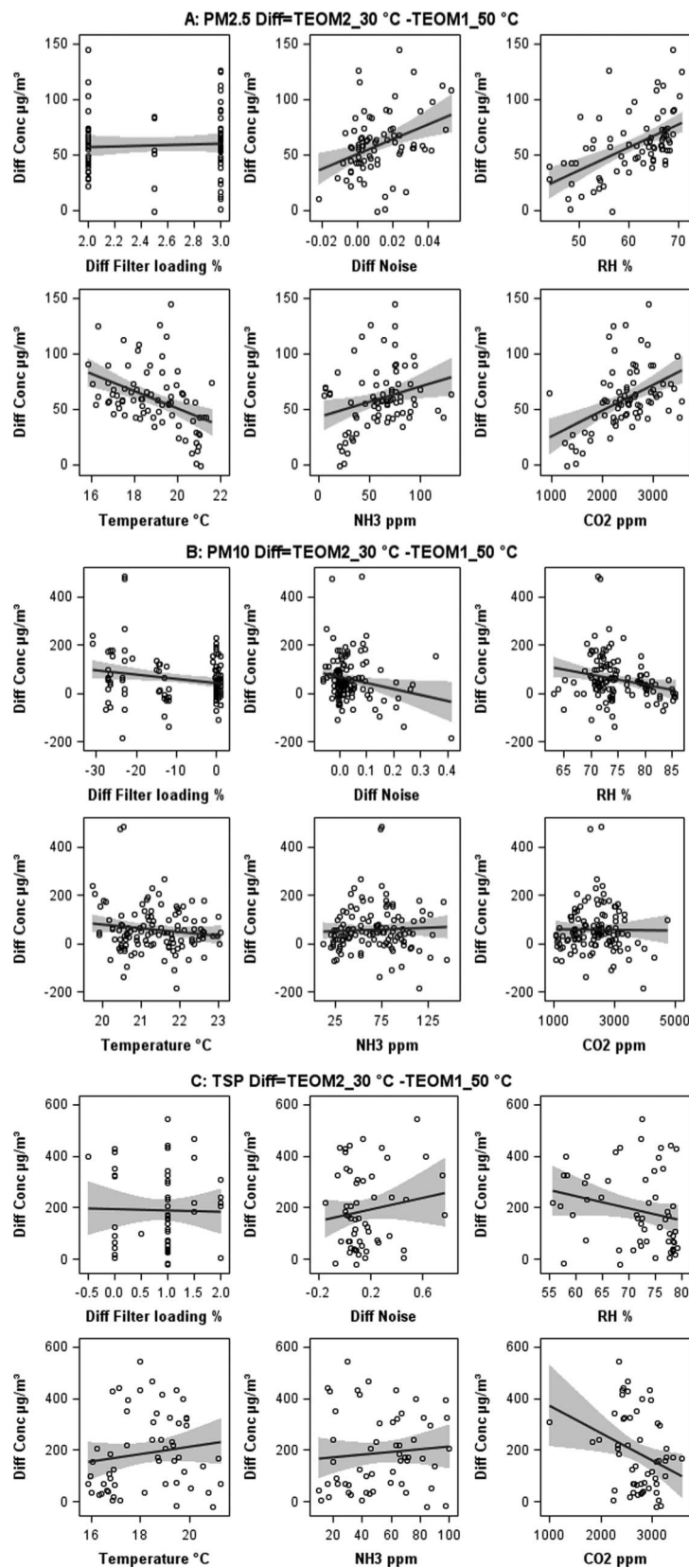


Figure 8. The relationships between the PM mass difference (Diff Conc = MC_{30 °C} – MC_{50 °C}) and the environmental and instrumental parameters (Diff Filter loading = filter loading difference, Diff Noise = noise difference, RH = relative humidity of surrounding air, temperature = in-house air temperature, NH₃ and CO₂) (shaded area = 95% CI).

Table 5. Parameter estimates for PM_{2.5}, PM₁₀, and TSP measurements*

	PM _{2.5} (R ² = 0.36)			PM ₁₀ (R ² = 0.15)			TSP (R ² = 0.17)		
	Parameter Estimate	P Value	Variance Inflation Factors	Parameter Estimate	P Value	Variance Inflation Factors	Parameter Estimate	P Value	Variance Inflation Factors
Intercept	−146	0.13	0	551	0.06	0	572	0.30	0
DiffPressDrop	2.36	0.71	1.35	−0.69	0.38	1.56	−15.1	0.68	1.23
Diff Noise	352	0.09	1.40	−312	0.001	1.11	55.2	0.62	1.24
RH	1.71	0.005	2.50	−4.05	0.02	1.40	−3.47	0.41	2.31
Temp	2.93	0.34	3.01	−8.43	0.44	1.69	10.8	0.58	2.31
NH ₃	−0.05	0.74	1.96	−0.34	0.44	4.15	−0.13	0.91	2.49
CO ₂	0.015	0.03	2.17	0.005	0.76	3.07	−0.12	0.02	1.09

Note: *Notations as in Table 4.

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